Highlights

Temperature-Dependent Nonlinear Resonance of a Bolted Lap Joint

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- Temperature changes both the magnitude and form of bolted-joint nonlinear resonance.
- The monolithic reference shifts to lower frequency at all 16 temperatures.
- Cold jointed backbones are non-monotonic and poorly described by a cubic model.
- A higher-order empirical model improves prediction across many temperatures.

Temperature-Dependent Nonlinear Resonance of a Bolted Lap Joint

J. B. Morris[a,*], M. S. Bonney[a] and H. H. Khodaparast[a]

[a]*Faculty of Science and Engineering, Swansea University, Fabian Way, Swansea, United Kingdom*

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ABSTRACT

Bolted joints can introduce amplitude-dependent dynamics that substantially modify the resonance behaviour of assembled structures. Although such nonlinearities are commonly characterised at a single environmental condition, many engineering structures operate over wide thermal ranges in which the mechanical state of the joint may vary simultaneously with vibration amplitude. This study experimentally investigates the temperature-dependent nonlinear resonance behaviour of a clamped–clamped bolted lap-joint beam using an equivalent monolithic beam as a reference configuration. Impact-hammer testing was followed by stepped-sine measurements at 16 nominal temperatures from −30 °C to 120 °C, with 16 excitation levels at each temperature, producing 256 resonance-peak observations per configuration. The monolithic specimen exhibited a negative backbone frequency shift at every thermal condition, ranging from −0.728 % to −0.331 %. In contrast, the jointed specimen changed from a net softening response at the coldest temperatures to positive backbone frequency shifts from 0 °C onward, with a maximum shift of 1.334 % at 20 °C. Linear interpolation of the model-independent backbone frequency shift gives a zero crossing at approximately −3.33 °C. A standard cubic Duffing-type backbone model captures the warmer jointed response but is poorly identified at −30 °C, −20 °C, and −10 °C, where the measured resonance-peak backbones are non-monotonic. An extended $A^2 + A^4$ empirical backbone representation improves leave-one-out predictive accuracy at many thermal conditions, demonstrating that a single cubic coefficient is not universally sufficient to describe the observed curvature. The results show that thermal state can alter both the magnitude and form of amplitude-dependent resonance behaviour in the jointed structure.

1. Introduction

Bolted joints are ubiquitous in assembled structures because they provide high load-carrying capacity while retaining practical advantages in manufacture, assembly, inspection, maintenance, and disassembly. Their dynamic behaviour is nevertheless difficult to predict because the load path through a joint is governed by contact pressure, friction, tangential compliance, preload, local slip, and assembly variability. These effects can produce amplitude-dependent stiffness and dissipation, non-smooth response characteristics, and appreciable uncertainty in the identified dynamic properties of otherwise nominally identical assemblies [1, 2]. Experimental benchmark studies have therefore placed increasing emphasis on controlled jointed structures in which nonlinear response, repeatability, and modelling assumptions can be assessed systematically [3, 4].

Temperature introduces a second source of variability in structural dynamics. Even in nominally continuous structures, changes in temperature can shift modal frequencies through temperature-dependent material properties, thermally induced stress, and boundary-condition effects [5, 6]. More generally, thermal-mechanical interactions can modify nonlinear dynamic behaviour when thermal and mechanical effects act concurrently rather than being treated as independent corrections [7, 8]. Recent experimental work on a thermally loaded bi-metallic beam has further demonstrated that temperature can alter nonlinear vibration characteristics substantially across a broad operating range [9]. These studies establish that environmental state can be an important part of nonlinear model identification, but they do not directly resolve how a discrete frictional joint modifies amplitude-dependent resonance over temperature.

Most experimental studies of bolted-joint dynamics have instead focused on a fixed environmental condition. Reported behaviours include amplitude-dependent resonance shifts, nonlinear stiffness, damping, and sensitivity to the state of the joint interface [10–13]. This body of work demonstrates that a joint cannot generally be represented by a single linear connection property over all response amplitudes. It also implies an important practical limitation: if the state of the interface changes with temperature, a nonlinear model identified at one thermal condition need not remain representative at another.

*Corresponding author
joshua.ben.morris@swansea.ac.uk (J.B. Morris)
ORCID(s): 0009-0001-8738-4312 (J.B. Morris); 0000-0002-1499-0848 (M.S. Bonney); 0000-0002-3721-4980 (H.H. Khodaparast)

A further question concerns how the measured nonlinearity should be parametrised. A convenient description of a resonance-peak backbone is the cubic Duffing-type relation

$$f_{\mathrm{res}}^2 = f_0^2 + \alpha A^2, \tag{1}$$

where the sign of α provides a compact indicator of an overall softening or hardening tendency [14]. This representation is useful when the resonance locus is approximately monotonic and well described by a single curvature trend. For a non-monotonic backbone, however, one fitted cubic coefficient can compress several amplitude-dependent trends into a single scalar and may therefore obscure changes in response form. Model adequacy must then be assessed alongside the fitted coefficient rather than inferred from its sign alone.

The present work experimentally examines this combined temperature–amplitude dependence using two clamped–clamped steel beam configurations: a single-bolt lap-joint specimen and an equivalent monolithic reference. Both configurations were subjected to the same thermal–mechanical stepped-sine procedure over 16 nominal temperatures and 16 excitation levels, yielding 256 resonance-peak observations for each configuration and 512 measured backbone points in total. The reference specimen is used to distinguish trends that are also present in the continuous structure from those that emerge only in the jointed configuration; because one specimen of each type was tested, the comparison is interpreted as a controlled mechanistic case study rather than population-level inference.

The present analysis makes three contributions. First, the measured backbone frequency shift is retained as the primary model-independent descriptor of net nonlinear resonance change across the temperature–amplitude dataset. Second, uncertainty in the fitted cubic coefficient is reported explicitly and interpreted together with the quality of the cubic representation, preventing weakly identified cold-temperature coefficients from being over-interpreted. Third, an empirical extended backbone model containing an additional A^4 term is assessed using leave-one-out cross-validation and the corrected Akaike information criterion, allowing additional model flexibility to be evaluated against predictive performance rather than training fit alone.

The study therefore addresses three questions:

(i) How does the amplitude-dependent resonance response of the jointed structure change with temperature relative to the monolithic reference?

(ii) Under which thermal conditions is a standard cubic Duffing-type backbone an adequate description of the measured resonance locus?

(iii) Does an additional empirical curvature term improve predictive description of the measured backbone once model complexity and out-of-sample prediction within each thermal condition are considered?

2. Experimental Methodology

2.1. Specimens and test procedure

Two steel beam configurations were investigated: a bolted specimen formed from two overlapping beam segments joined by a single lap-joint fastener, and an equivalent monolithic reference with the same overall structural arrangement but no mechanical interface.

Both specimens were mounted in the same clamped–clamped fixture on the shaker assembly inside an environmental chamber. Figure 1 summarises the measurement and excitation chain and the two test configurations. The end clamps were tightened to 3 N m using a calibrated torque wrench and were not readjusted during testing. The lap joint incorporated an M5 fastener tightened to 1.5 N m. Four response accelerometers were distributed along the beam span, with an additional accelerometer mounted on the base plate.

The physical specimen masses were

$$m_M = 0.228\,210\,\mathrm{kg}, \qquad m_J = 0.233\,040\,\mathrm{kg}. \tag{2}$$

Impact-hammer testing was performed before the stepped-sine campaign. The identified first resonance occurred at 150.374 752 Hz for the monolithic specimen and 116.937 585 Hz for the jointed specimen.

For a compact baseline comparison, a physical-mass-based stiffness-related estimate was calculated as

$$K^* = m(2\pi f_n)^2. \tag{3}$$

This gives 203.725 kN m^{-1} for the monolithic specimen and 125.805 kN m^{-1} for the jointed specimen, corresponding to a reduction of approximately 38.25 %. Because physical specimen mass is used rather than an identified modal mass, K^* is interpreted only as a stiffness-related comparison and not as modal stiffness.

For stepped-sine testing, each specimen was excited harmonically over an approximately 5 Hz-wide interval containing the first resonance. The commanded frequency was incremented in nominal 0.1 Hz steps. At each dwell, acceleration data were acquired at 1000 Hz for 2000 samples, corresponding to a 2 s record. Testing was performed at 16 nominal chamber temperatures,

$$T = -30, -20, \ldots, 120^{\circ}\mathrm{C},$$

with 16 commanded excitation levels from 0.5 to 8 in increments of 0.5. These drive settings define the test sequence and are not treated as measured response amplitudes.

Before each temperature-dependent sweep, the chamber was allowed to stabilise until the chamber thermocouple remained within approximately ±0.5 °C of the target set point for at least 10 minutes. The full frequency sequence was then repeated at each excitation level before proceeding to the next nominal temperature. This produced 256 resonance observations per specimen and 512 extracted backbone points in total.

The nominal chamber set point is used as the thermal coordinate throughout the analysis.

2.2. Signal processing and resonance extraction

Temperature-dependent accelerometer sensitivity compensation was applied upstream in the LabVIEW acquisition workflow using a third-order polynomial derived from manufacturer calibration information. MATLAB post-processing therefore operated on acceleration signals for which this temperature correction had already been applied.

For the stepped-sine measurements, the four response accelerometer channels were processed individually and then averaged to form the system-level response measure used for resonance extraction. Each channel was de-trended and high-pass filtered with a second-order Butterworth filter at 35 Hz, applied using zero-phase forward–reverse filtering. The harmonic component at the commanded excitation frequency was then extracted for each dwell. The commanded drive setting was used only to define the test sequence and not as a response coordinate.

At each excitation level, the processed response was assembled as a discrete function of excitation frequency. The maximum measured response was first identified on the experimental frequency grid, after which the peak location was refined using piecewise cubic Hermite interpolation (PCHIP). This interpolation refines the numerical peak-frequency estimate but does not add independent experimental information between measured frequency points. The resulting loci are therefore described throughout as forced-response resonance peaks rather than nonlinear-normal-mode backbones.

The response displacement amplitude was reconstructed from the harmonic acceleration component as

$$A = \frac{a}{(2\pi f)^2}, \tag{4}$$

where a is the harmonic acceleration amplitude expressed in physical units and f is the associated excitation frequency. In the processing workflow, acceleration measured in units of g is converted to m s^{-2} before applying Eq. (4). The resulting quantity is a harmonic displacement estimate rather than a directly measured displacement or modal coordinate. All backbone models use this reconstructed response amplitude.

3. Nonlinear Identification

3.1. Backbone metrics and cubic model

The model-independent backbone frequency shift was defined as

$$\mathrm{BFS} = \frac{f_{\mathrm{high}} - f_{\mathrm{low}}}{f_{\mathrm{low}}} \times 100\%. \tag{5}$$

A negative value represents a net downward resonance shift with increasing amplitude, while a positive value represents a net upward shift.

The standard phenomenological model was Eq. (1). The coefficient α is reported in Hz^2 µm^{-2}. It is a backbone coefficient, not a physical cubic restoring-force coefficient in N m^{-3}. For each fit, R^2 and conditional 95 % least-squares confidence intervals were calculated.

Table 1
Baseline dynamic properties of the two specimens.

Specimen	Physical mass (kg)	Impact resonance (Hz)	K^* (kN/m)
Monolithic	0.22821	150.375	203.725
Jointed	0.23304	116.938	125.805

$K^* = m(2\pi f_n)^2$ uses physical specimen mass and is therefore a stiffness-related estimate rather than a formally identified modal stiffness.

3.2. Extended-model comparison and robustness

The extended empirical model was

$$f_{\mathrm{res}}^2 = f_0^2 + \alpha A^2 + \beta_4 A^4. \tag{6}$$

The A^4 term is interpreted as an empirical curvature term rather than evidence of a physical fifth-order restoring force.

Predictive performance was assessed using leave-one-out cross-validation (LOOCV). Model complexity was assessed using

$$AIC_c = n \ln\left(\frac{RSS}{n}\right) + 2k + \frac{2k(k+1)}{n-k-1}. \tag{7}$$

The comparison quantity was

$$\Delta AIC_c = AIC_{c,\mathrm{standard}} - AIC_{c,\mathrm{extended}}. \tag{8}$$

The pipeline classified the extended model as preferred when its LOOCV RMSE was at least 5 % lower and its AIC_c at least 2 lower than the standard model. This is an exploratory screening rule rather than a formal hypothesis test.

For the extended model, a non-zero stationary point is

$$A_{\mathrm{turn}} = \sqrt{-\frac{\alpha}{2\beta_4}} \tag{9}$$

when the argument is positive. Residual-bootstrap resampling was used to assess conditional fitting sensitivity. Peak-estimator sensitivity was separately checked using raw-grid, PCHIP, and local-quadratic peak localisation.

4. Results

4.1. Linear and measured backbone response

The monolithic specimen exhibited a first impact-hammer resonance at 150.375 Hz, whereas the jointed specimen occurred at 116.938 Hz. The physical-mass-based stiffness estimate decreases from 203.725 kN m^{-1} to 125.805 kN m^{-1}, a relative reduction of 38.248 %.

Baseline quantities are summarised in Table 1.

Figure 3 shows the overall temperature evolution of representative measured resonance-peak backbones, normalised by the lowest-amplitude resonance at each temperature. The monolithic specimen exhibits a consistent downward frequency shift with increasing amplitude, whereas the jointed specimen changes markedly with thermal state, from a cold non-monotonic response to a pronounced positive shift at warmer temperatures. Open circles denote measured resonance-peak observations; the solid curves are MAKIMA interpolants included only as visual guides between measured points.

The configuration contrast is particularly clear at 20 °C, as shown in Fig. 4. Over the measured amplitude range, the monolithic resonance decreases monotonically while the jointed resonance increases, providing a direct matched-condition illustration of the opposite net nonlinear trends.

The cold jointed measurements are isolated in Fig. 5. At −30 °C, −20 °C, and −10 °C, the resonance locus initially decreases with amplitude before flattening or recovering. By 0 °C, the endpoint shift is positive, although the measured locus retains curvature that is not captured well by a single cubic trend.

Table 2
Representative temperature-dependent nonlinear results. The cubic coefficient is reported in units of $10^{-4}\,\mathrm{Hz}^2/\mu\mathrm{m}^2$. Confidence intervals are conditional least-squares intervals for the fitted backbone model.

T (°C)	BFS_M (%)	BFS_J (%)	α_M	R^2_M	α_J	95% CI_J	R^2_J
-30	-0.728	-0.616	-4.695	0.828	-0.643	[-1.638, 0.352]	0.121
-20	-0.465	-0.176	-2.918	0.906	0.170	[-0.400, 0.739]	0.028
-10	-0.530	-0.352	-3.242	0.968	0.228	[-0.388, 0.844]	0.043
0	-0.530	0.176	-3.625	0.959	0.927	[0.664, 1.189]	0.803
20	-0.728	1.334	-5.155	0.973	2.283	[1.496, 3.071]	0.734
50	-0.597	1.065	-4.560	0.947	2.688	[1.897, 3.479]	0.791
120	-0.462	0.440	-5.515	0.970	2.374	[1.965, 2.782]	0.917

4.2. Temperature dependence of BFS and Duffing coefficient

The monolithic BFS remains negative across the entire thermal range, between approximately -0.7284 % and -0.3312 %. The jointed specimen changes from -0.6163 % at -30 °C to positive BFS by 0 °C. Linear interpolation between -10 °C and 0 °C gives

$$T_{\mathrm{BFS}=0} \approx -3.33\,^{\circ}\mathrm{C}. \tag{10}$$

The maximum measured jointed BFS is 1.3339 % at 20 °C.

The monolithic α remains negative across the complete thermal range. The jointed cold-temperature fits are poorly identified: at -30 °C, -20 °C, and -10 °C, R^2 values are approximately 0.121, 0.028, and 0.043, respectively, and the conditional 95 % intervals for α include zero.

At 0 °C,

$$\alpha_J = 9.27 \times 10^{-5}\ \mathrm{Hz}^2/\mu\mathrm{m}^2, \qquad R^2 = 0.803,$$

with the conditional interval entirely above zero. At 20 °C,

$$\alpha_J = 2.28 \times 10^{-4}\ \mathrm{Hz}^2/\mu\mathrm{m}^2, \qquad R^2 = 0.734.$$

Representative values are summarised in Table 2.

4.3. Standard versus extended model performance

At -30 °C, the standard model cannot reproduce the measured decrease and recovery in resonance frequency. The extended model reduces LOOCV RMSE from 0.2479 Hz to 0.2130 Hz. At 20 °C, LOOCV RMSE decreases from 0.2389 Hz to 0.1956 Hz.

The extended representation passes the exploratory screening criterion at 14/16 monolithic temperatures and 13/16 jointed temperatures. The improvement produced by the A^4 term is therefore not specific to the jointed configuration.

Representative model-comparison values are listed in Table 3. The cold jointed extended-model stationary points are shown in their fitting context in Fig. 9. At -30 °C and -20 °C, the extended model passes the exploratory screening and gives model-derived stationary-point amplitudes of approximately 0.677 mm and 0.637 mm, respectively. At -10 °C, a stationary point can still be computed from the extended fit, but the standard model is preferred by the screening criterion; that stationary point is therefore retained only as a conditional model output rather than a preferred description of the measured response. Peak-estimator sensitivity was additionally assessed using raw-grid, PCHIP, and local-quadratic peak localisation; the resulting cold-jointed stationary-point estimates showed the same temperature ordering and only small numerical variation, indicating that the trend was not created by a single peak-localisation method.

5. Discussion

5.1. Temperature-dependent nonlinear behaviour

The most robust experimental observation is the contrast between the two measured backbone families. The monolithic structure exhibits a negative BFS at every tested temperature. The jointed specimen shows non-monotonic

Table 3
Representative model-validation results for the jointed specimen. LOOCV values are RMSE in Hz; the screening decision is exploratory rather than a formal hypothesis test.

T (°C)	Std. LOOCV	Ext. LOOCV	Improvement (%)	ΔAIC_c	Preferred
-30	0.2479	0.2130	14.08	5.35	Extended
-20	0.1615	0.1464	9.37	2.39	Extended
-10	0.1910	0.1952	-2.19	-0.62	Standard
20	0.2389	0.1956	18.13	7.58	Extended

cold backbones and negative endpoint BFS, followed by a positive net shift from 0 °C onward. These statements are direct descriptions of the measured forced-response peak loci and do not require assignment of a specific contact mechanism.

The model-independent BFS zero crossing occurs between -10 °C and 0 °C, with a linear interpolation estimate near -3.33 °C. This estimate is more defensible than assigning a precise transition temperature from the sign of the cubic coefficient because the cold cubic fits have low R^2 and conditional confidence intervals that span zero. The interpolated value should therefore be interpreted as a descriptive crossover in the endpoint backbone metric rather than as a uniquely identified physical transition temperature.

BFS uses only the first and final resonance points, whereas α is estimated from all measured points in the transformed $(A^2, f_{\mathrm{res}}^2)$ domain. For a non-monotonic backbone, one fitted slope compresses multiple local trends into a single scalar quantity. The poor cold jointed R^2 values therefore indicate that the assumed cubic representation is inadequate at those conditions, rather than providing strong evidence for a well-defined change in the sign of a physical cubic stiffness parameter.

The extended empirical model improves the representation of curvature at many conditions. However, because it is selected at a similarly high fraction of monolithic and jointed temperatures, the A^4 contribution should not be interpreted as a unique contact-mechanics parameter or as evidence of a physical fifth-order restoring force. Instead, it demonstrates that many measured resonance backbones contain curvature beyond that captured by a single A^2 term. The model-selection rule is exploratory and the LOOCV results constitute internal predictive assessment rather than validation against an independent experimental dataset.

The cold-jointed stationary-point analysis in Fig. 9 provides an additional model-based description of this curvature, but it does not alter that interpretation. In particular, the -10 °C stationary point arises from an extended model that does not pass the screening criterion, while the -30 °C and -20 °C estimates remain conditional on the fitted empirical form and residual-bootstrap procedure.

5.2. Modelling implications and physical interpretation

The configuration-level comparison shows that the jointed specimen develops a temperature-dependent nonlinear evolution that is qualitatively absent from the monolithic reference. Bolted interfaces are well known to introduce non-smooth nonlinear behaviour, uncertainty, frictional energy dissipation, and preload-sensitive dynamic properties [1, 10, 13]. Classical tangential-contact theory also establishes that contact compliance and the distribution of stick and slip depend on normal loading and tangential excitation [15, 16]. These ideas make temperature-dependent contact-state evolution, tangential stiffness, frictional slip, and effective preload plausible contributors to the measured jointed response.

The present measurements do not, however, directly identify any one of these mechanisms. Bolt load, interface pressure, local relative motion, and slip state were not measured, and temperature can also alter structural frequencies through material, stress, and boundary-condition effects even in structures without a discrete joint [5, 6, 8]. The physical interpretation should therefore remain that the observed trends are *consistent with* a temperature-dependent change in the joint/contact state superimposed on structural thermal effects, rather than proof of a specific microslip-hardening or preload mechanism.

A model calibrated at 20 °C would identify the jointed system as strongly hardening, while at -30 °C the same specimen exhibits a net negative BFS and a non-monotonic resonance locus. A temperature-independent nonlinear parameter set would therefore fail to describe this specimen across the tested range. In practical terms, a room-temperature model could predict the wrong direction of amplitude-dependent resonance movement under cold

operation. This is relevant whenever resonance proximity controls dynamic amplification, response margins, or model-based qualification across a thermal operating envelope.

Moreover, the cold failure of the cubic model indicates that defining only $\alpha = \alpha(T)$ may not be sufficient; the functional form of the amplitude dependence may itself require additional flexibility with temperature. The fact that the A^4 representation also improves prediction for most monolithic conditions reinforces that this curvature should be treated as a general empirical modelling feature rather than a unique signature of joint contact. Consequently, the journal contribution is not the identification of a new contact-law coefficient, but the experimental demonstration that both the magnitude and the descriptive model form of the forced-response resonance behaviour vary with thermal state.

5.3. Limitations

Only one specimen of each configuration was tested, so specimen-to-specimen, assembly, and reassembly variability are not quantified. The comparison is therefore a controlled case study and should not be interpreted as population-level inference. Although the monolithic specimen provides an important reference, differences between the two configurations necessarily include the joint geometry and load path and may also include associated differences in modal properties and effective mass; the observed contrast cannot be partitioned uniquely into individual contact mechanisms.

Bolt preload, interface pressure, local relative motion, and slip were not measured directly. The backbones are forced-response peak loci rather than direct nonlinear normal-mode measurements and can therefore depend on damping, forcing level, sweep protocol, jumps, and the finite frequency grid. The 0.1 Hz excitation spacing limits experimental peak-frequency resolution, while PCHIP interpolation only refines the numerical peak estimate and does not add independent experimental information. The conditional least-squares intervals for α, the residual-bootstrap intervals, and the peak-estimator sensitivity check quantify fitting and numerical-extraction sensitivity for the available backbone points; they do not represent repeatability uncertainty or a complete measurement-uncertainty budget. Finally, the extended-model screening and LOOCV assessment are internal to the same dataset and should be regarded as model-adequacy diagnostics rather than independent experimental validation.

6. Conclusions

The temperature-dependent nonlinear resonance behaviour of a bolted lap-joint beam was experimentally characterised relative to an equivalent monolithic reference.

The first impact-hammer resonance decreases from approximately 150.375 Hz for the monolithic specimen to 116.938 Hz for the jointed configuration, while the associated physical-mass-based stiffness estimate decreases by approximately 38.25 %. The monolithic specimen exhibits a negative backbone frequency shift at all 16 temperatures. The jointed specimen changes from a net softening response at the coldest conditions to a positive backbone shift from 0 °C onward, with a BFS zero crossing estimated at approximately −3.33 °C and a maximum measured BFS of approximately 1.334 % at 20 °C.

The standard cubic Duffing-type model is poorly identified for the jointed backbones at −30 °C, −20 °C, and −10 °C, while from 0 °C upward the fitted cubic coefficient is positive and substantially better identified. An extended $A^2 + A^4$ empirical backbone model improves predictive performance at many thermal conditions, but because it is also preferred for most monolithic conditions, the additional curvature term should be regarded as a general empirical modelling improvement rather than a unique contact-mechanics parameter.

The central experimental conclusion is that thermal state can alter both the magnitude and the form of the amplitude-dependent resonance behaviour of the jointed structure. A nonlinear model identified at a single thermal condition cannot be assumed to remain valid throughout a broad operating envelope.

Future work should investigate repeat assemblies, direct bolt-load or interface-force measurement, different preload levels, multi-bolt configurations, and higher-order modes.

Nomenclature

A	reconstructed harmonic displacement amplitude
A_{turn}	model-derived stationary-point amplitude
BFS	backbone frequency shift
f_0	fitted zero-amplitude backbone-frequency intercept
f_{low}	resonance-peak frequency at the lowest valid response amplitude
f_{high}	resonance-peak frequency at the highest valid response amplitude
f_{res}	measured forced-response resonance-peak frequency
K^*	physical-mass-based stiffness-related estimate, $m(2\pi f_n)^2$
m	physical specimen mass
T	nominal chamber temperature
α	fitted cubic backbone coefficient
β_4	fitted fourth-order amplitude coefficient in the extended backbone model
RSS	residual sum of squares
AIC_c	corrected Akaike information criterion
ΔAIC_c	$AIC_{c,\text{standard}} - AIC_{c,\text{extended}}$
n	number of observations used in a model fit
k	number of fitted model parameters
$\text{RMSE}_{\text{LOOCV}}$	leave-one-out cross-validation root-mean-square error

CRediT authorship contribution statement

J. B. Morris: Conceptualization, Methodology, Investigation, Formal analysis, Software, Visualization, Original draft. **M. S. Bonney:** Supervision, Reviewing and editing. **H. H. Khodaparast:** Supervision, Reviewing and editing.

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Data Availability

The experimental data and MATLAB analysis code supporting the findings of this study are available from the Swansea University Thermal-Mechanical Nonlinearity Testing (SUTMNT) repository:

https://github.com/joshuabmorris/SUTMNT/.

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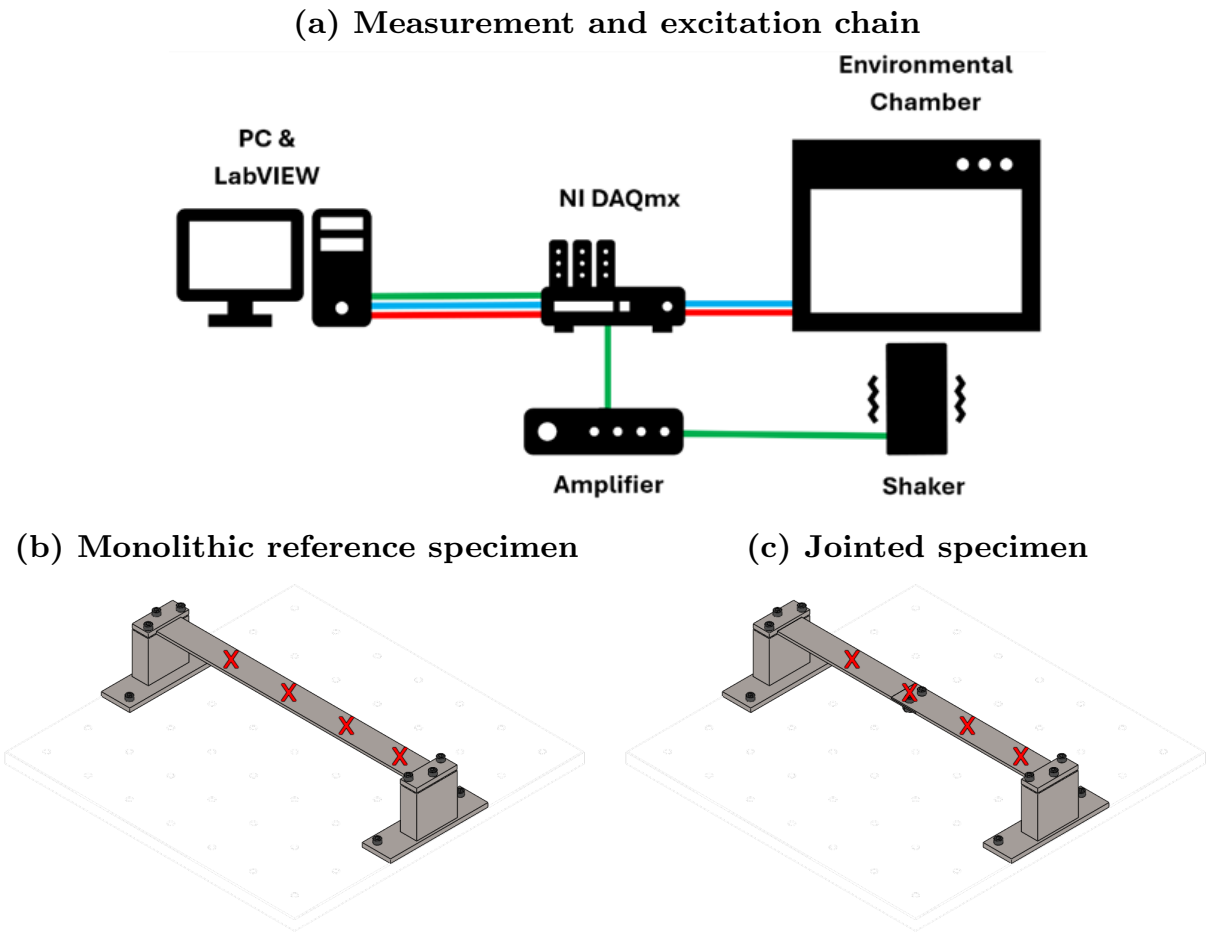


Figure 1: Experimental arrangement and test configurations. (a) Measurement and excitation chain linking the PC/LabVIEW system, NI DAQ, amplifier, shaker, and environmental chamber. (b) Monolithic reference specimen in the common clamped–clamped fixture. (c) Jointed specimen with the single overlapping bolted lap joint. Red crosses indicate the four span— response-accelerometer locations; the base accelerometer is omitted from panels (b) and (c) for clarity.

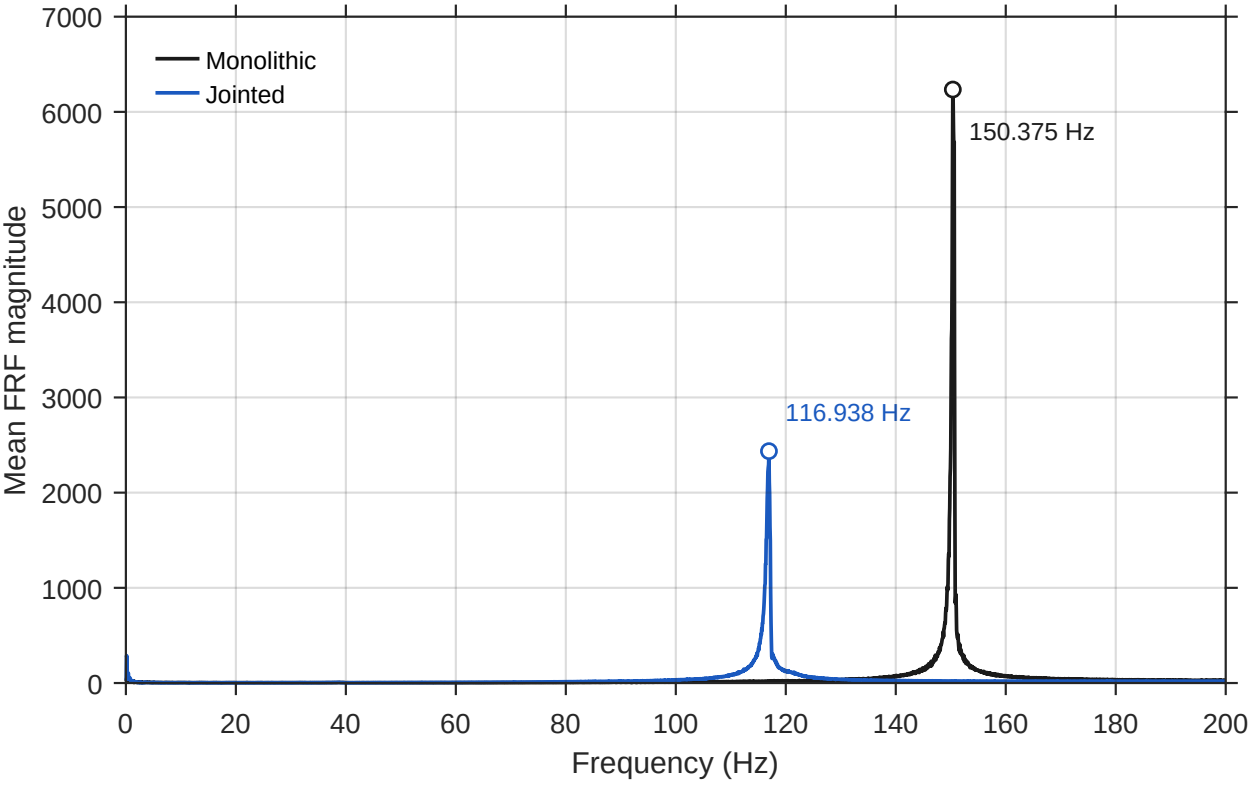


Figure 2: Impact-hammer response comparison for the monolithic and jointed specimens. Curves show the mean measured FRF magnitude, and open circles mark the identified first-resonance peaks.

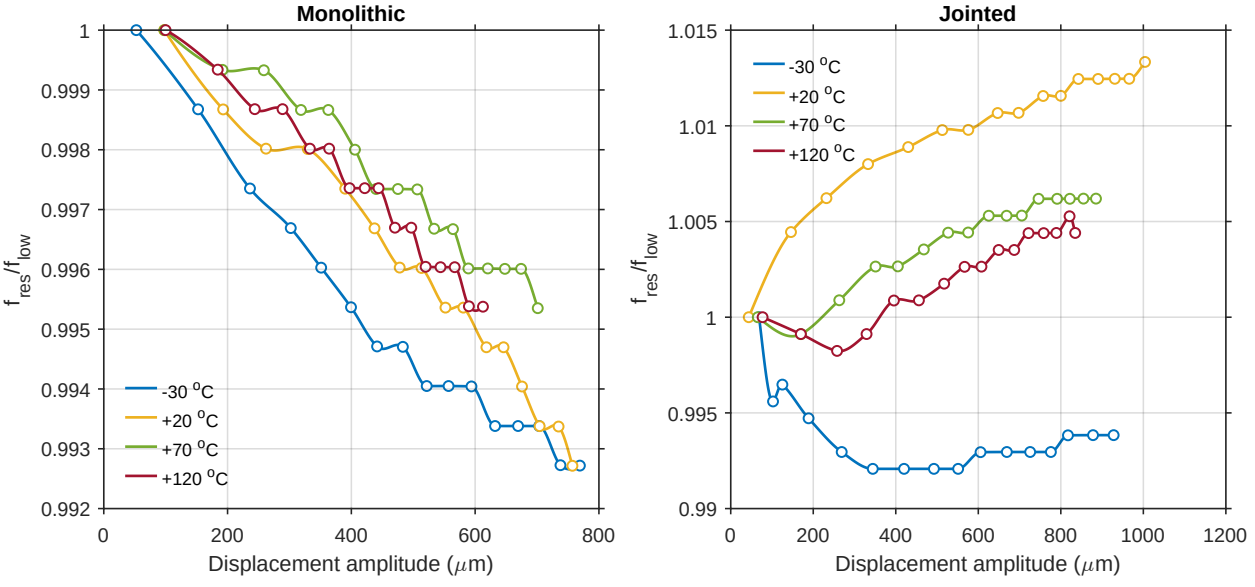


Figure 3: Representative temperature-dependent resonance-peak backbone evolution for the monolithic and jointed specimens. Resonance frequency is normalised by the lowest-amplitude resonance at each temperature. Open circles denote measured resonance-peak observations; solid curves are MAKIMA interpolants used only as visual guides between measured points and are not fitted physical models.

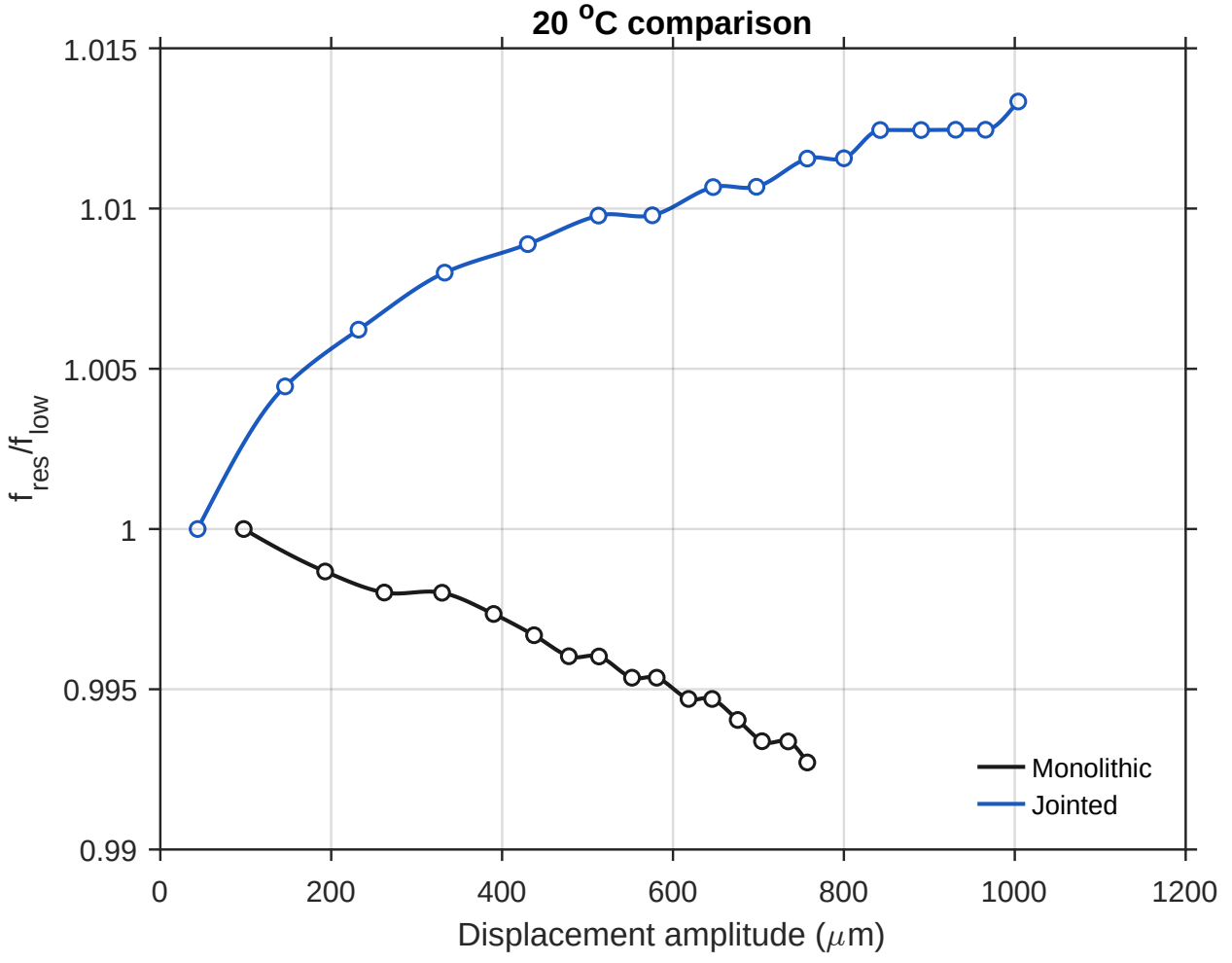


Figure 4: Normalised monolithic and jointed resonance-peak backbones at 20 °C. Open circles denote measured resonance-peak observations; solid curves are MAKIMA visual guides only.

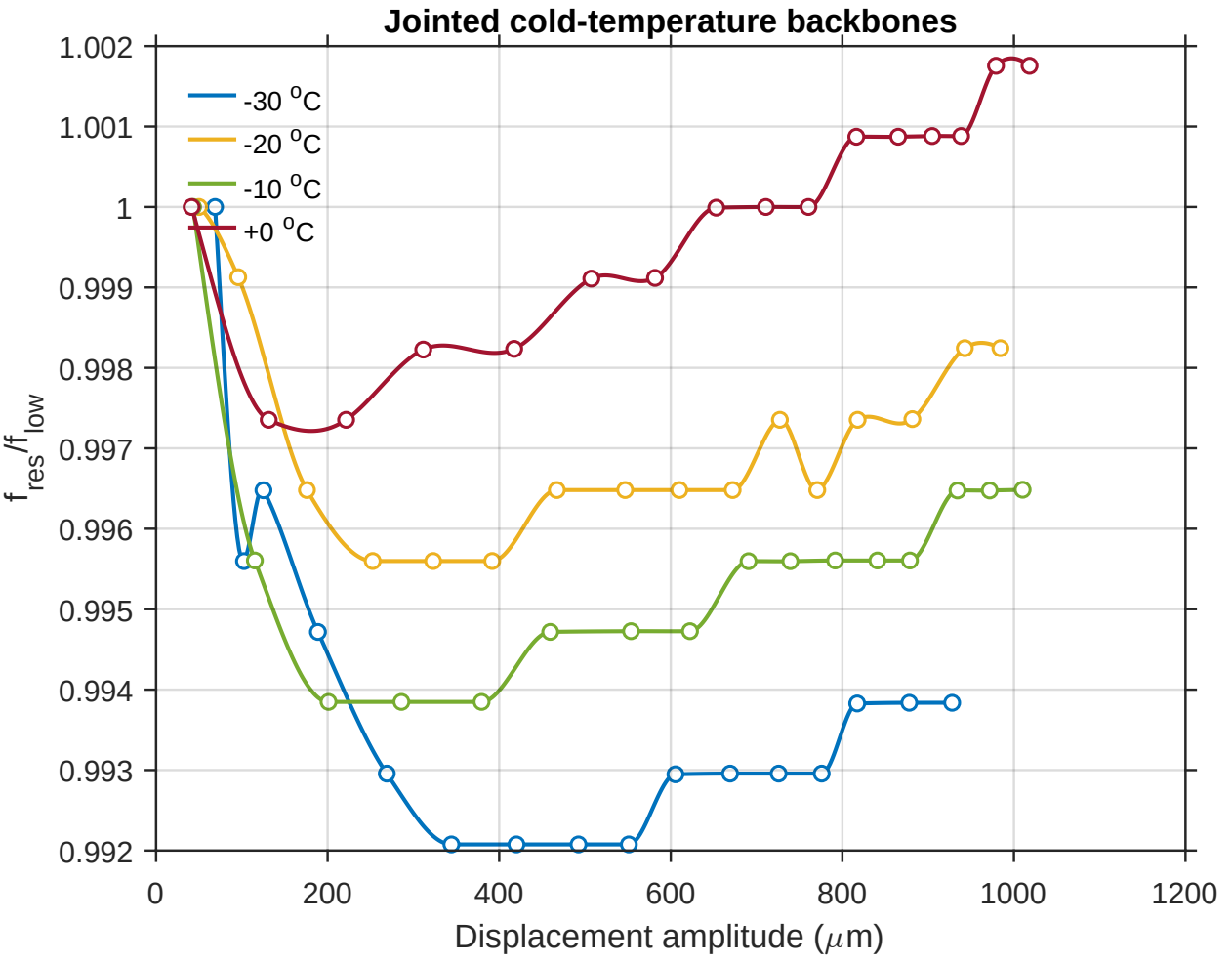


Figure 5: Measured jointed resonance-peak backbones at the coldest thermal conditions. Resonance frequency is normalised by the lowest-amplitude resonance at each temperature. Open circles denote measured observations; solid curves are MAKIMA visual guides only.

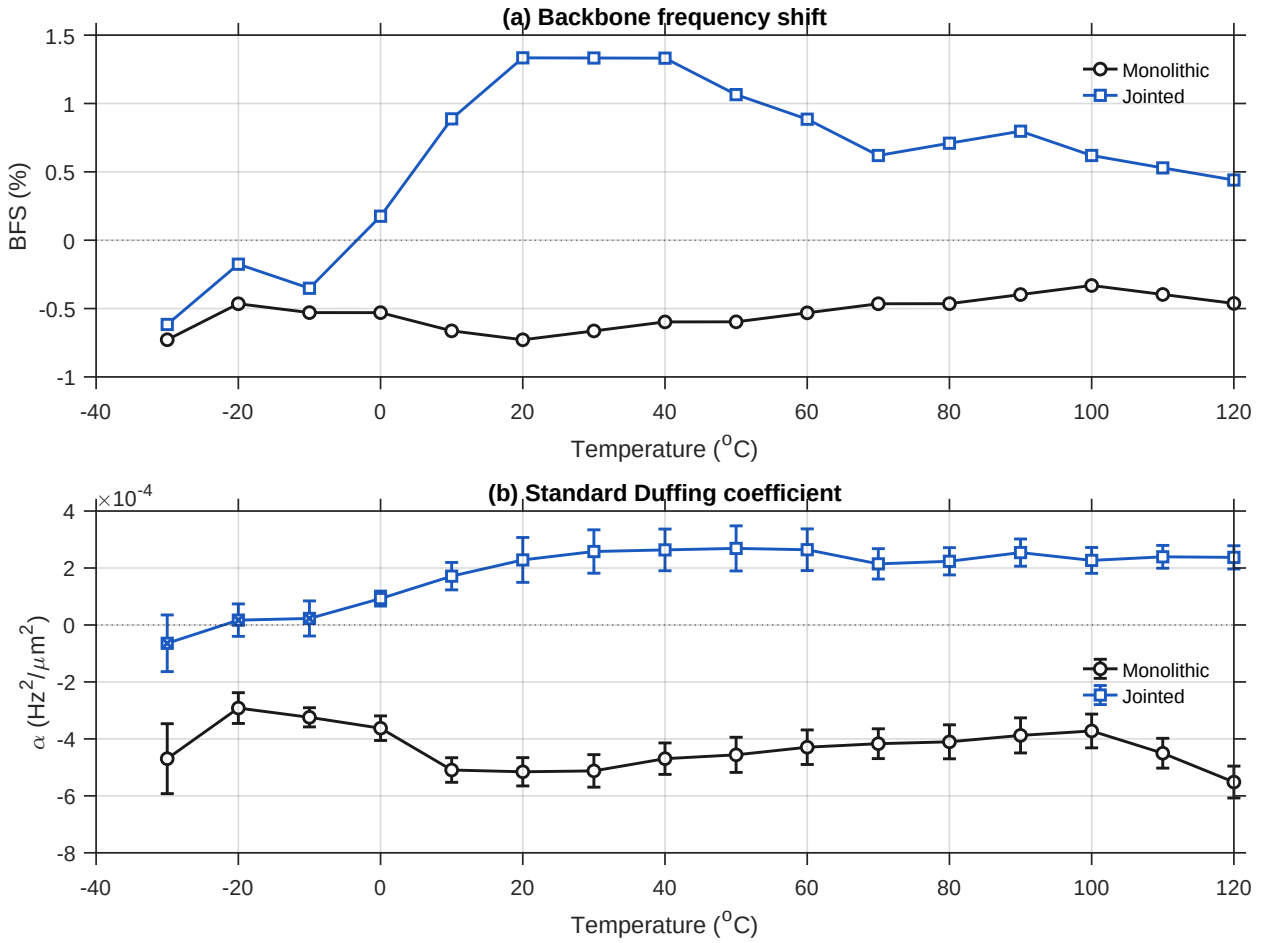


Figure 6: Temperature dependence of global nonlinear metrics. (a) Backbone frequency shift. (b) Standard cubic Duffing-type coefficient α with conditional 95 % least-squares intervals. Crosses identify standard-model fits with $R^2 < 0.5$.

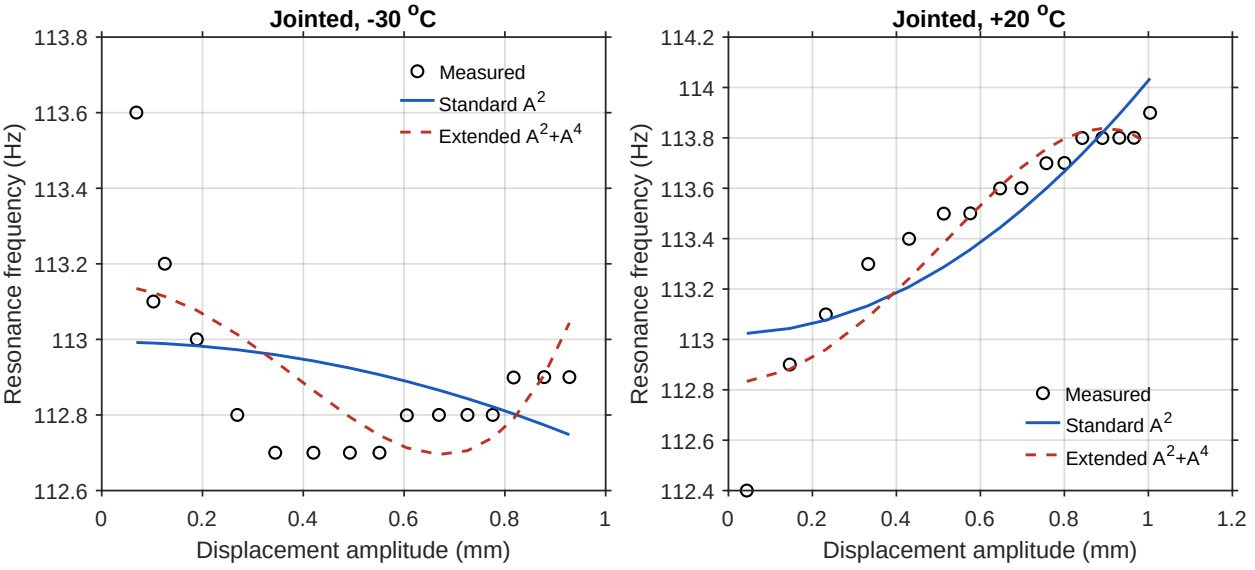


Figure 7: Representative standard and extended empirical backbone fits for the jointed specimen at −30 °C and 20 °C. Open circles denote measured resonance-peak observations; solid blue curves show the standard A^2 model and dashed red curves show the extended $A^2 + A^4$ model.

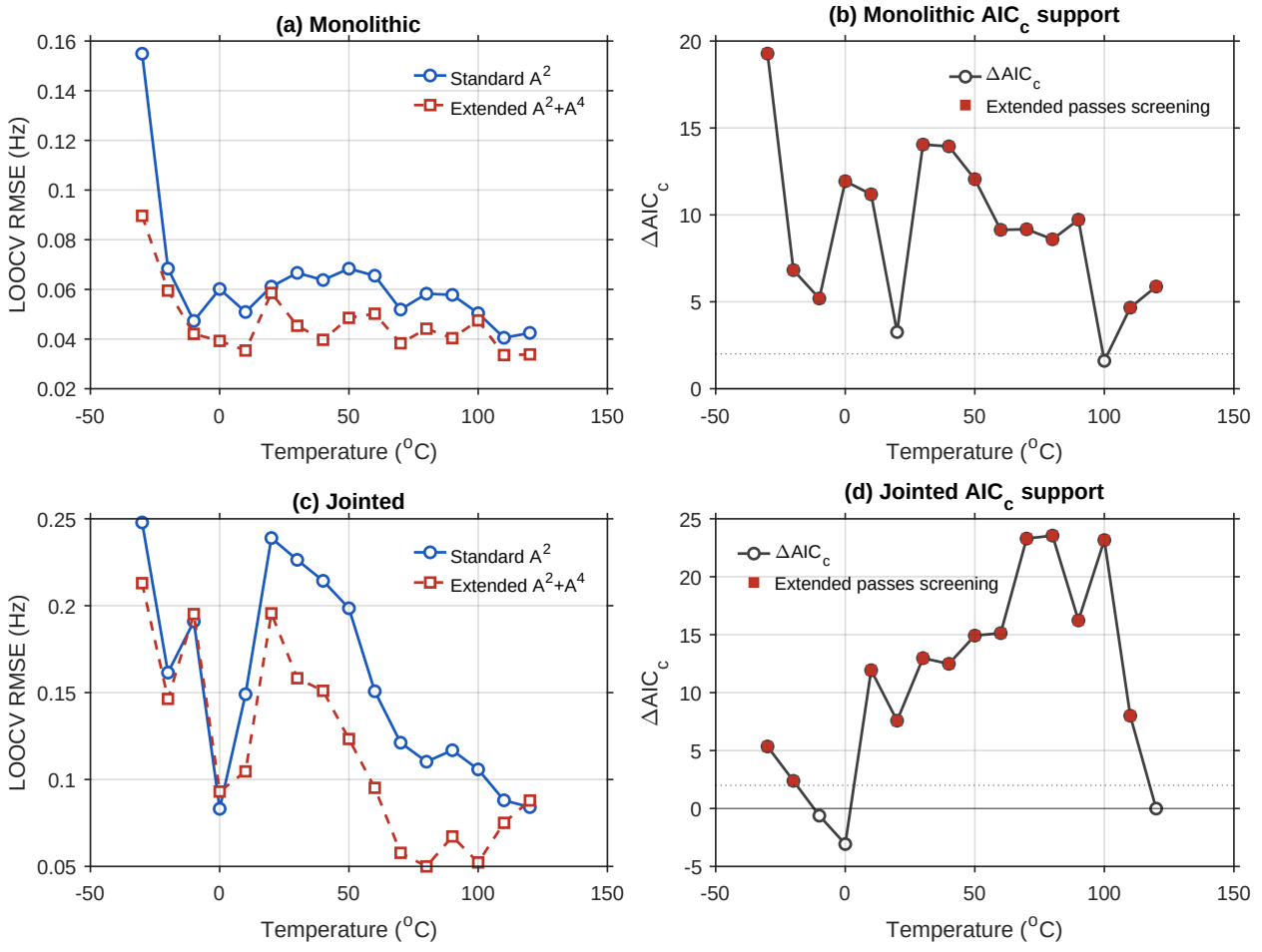


Figure 8: Predictive and complexity-based comparison of standard and extended backbone models across temperature. Left panels show LOOCV RMSE. Right panels show $\Delta AIC_c = AIC_{c,\mathrm{std}} - AIC_{c,\mathrm{ext}}$; the dotted horizontal line marks $\Delta AIC_c = 2$, one component of the exploratory screening rule. Filled red squares identify temperatures at which the extended model passes the complete screening criterion.

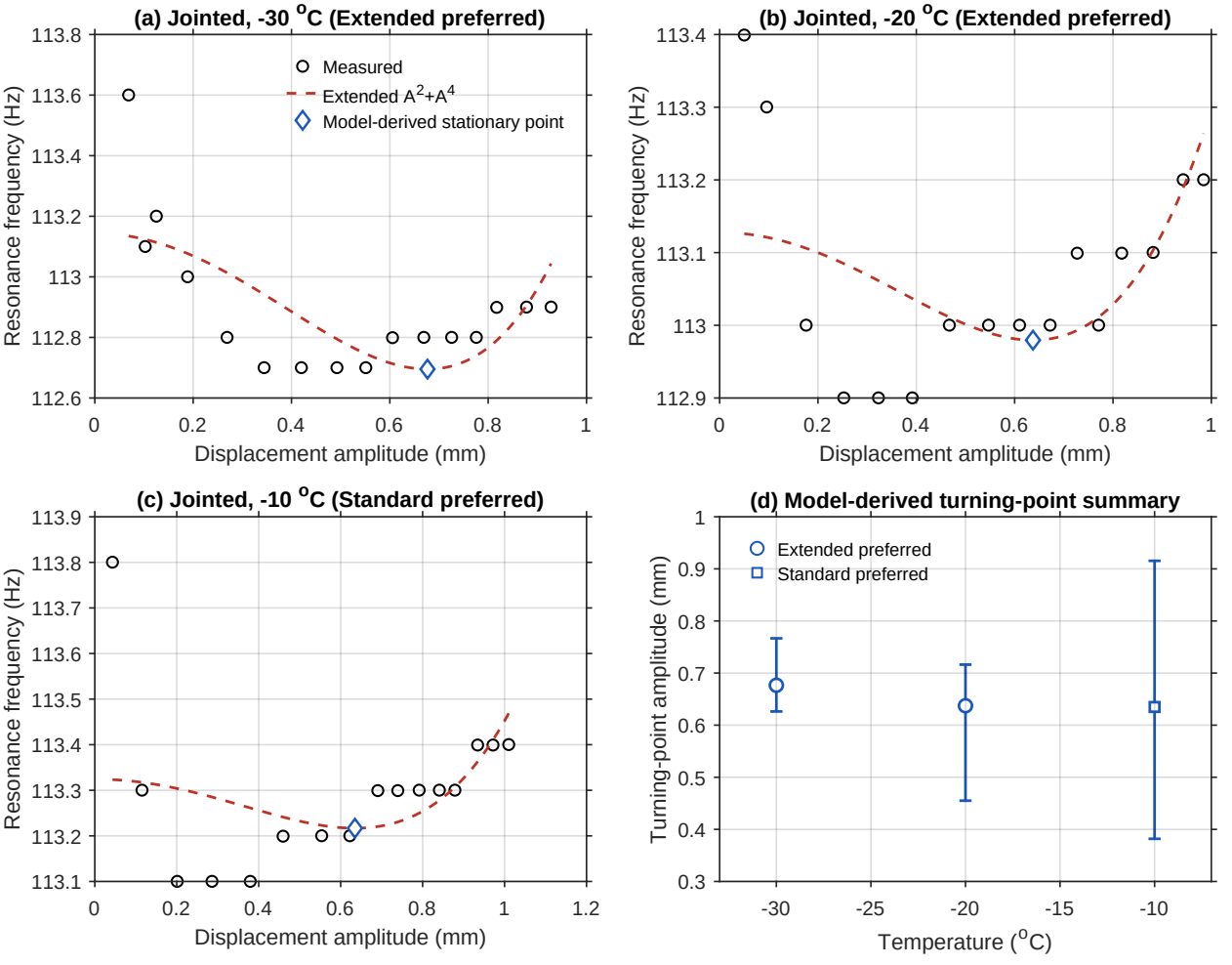


Figure 9: Model-derived cold-jointed stationary-point context. Panels (a)–(c) show the measured jointed resonance-peak observations, the extended $A^2 + A^4$ fit, and the stationary point derived from that fit at −30 °C, −20 °C, and −10 °C. The panel titles state whether the extended or standard model is preferred by the exploratory screening criterion. Panel (d) summarises the stationary-point amplitudes with conditional residual-bootstrap intervals; circles indicate temperatures at which the extended model is preferred and the square indicates the −10 °C condition where the standard model is preferred. These stationary points are model-derived quantities and are not interpreted as directly observed contact transitions.